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# Challenges in classifying cavitation: Correlating high-speed optical imaging and passive acoustic mapping of cavitation dynamics

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## ABSTRACT:

Both the biological effects and acoustic emissions generated by cavitation are functions of bubble dynamics. Monitoring of acoustic emissions is therefore desirable to improve treatment safety and efficacy. The relationship between the emission spectra and bubble dynamics is, however, complex. The aim of this study was to characterise this relationship for single microbubbles using simultaneous ultra-high-speed optical imaging and passive acoustic mapping of cavitation emissions. As expected, both the number of discrete harmonics and broadband content in the emissions increased with increasing amplitude of bubble oscillation, but the spectral content was also dependent upon other variables, including the frequency of bubble collapse and receiving transducer characteristics. Moreover, phenomena, such as fragmentation and microjetting, could not be distinguished from spherical oscillations when using the full duration acoustic waveform to calculate the emission spectra. There was also no correlation between the detection of broadband noise and widely used thresholds for distinguishing bubble dynamics. It is therefore concluded that binary categorisations, such as stable and inertial cavitation, should be avoided, and different types of bubble behavior should not be inferred on the basis of frequency content alone. Treatment monitoring criteria should instead be defined according to the relevant bioeffect(s) for a particular application.

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## I. INTRODUCTION

Acoustic cavitation offers substantial potential benefits in multiple applications of therapeutic ultrasound. These include enhanced thermal and mechanical ablation of tissue and improved efficiency of drug delivery (Kooiman *et al.*, 2020; Xu *et al.*, 2021; Zhang *et al.*, 2015). These effects are mediated by a range of physical phenomena produced by microbubble oscillations, such as microstreaming, microjetting, and shock wave generation. As they oscillate, bubbles produce acoustic emissions, the amplitude of which increases with the amplitude of oscillation. At the pressures typically used in ultrasound therapy, these emissions can be detected from outside the body to enable real time treatment monitoring using similar hardware to that used in ultrasound imaging. This is highly desirable to ensure that cavitation is occurring at the desired location.

A secondary goal of treatment monitoring is to relate the characteristics of the acoustic emissions to the bubble dynamics and ultimately to the likelihood of the desired (or undesired) bioeffects. Achieving this goal, however, is by no means straightforward. First, it is unclear in many cases precisely which phenomena are responsible for particular

bioeffects. Second, despite the widespread use of spectral features, such as harmonics, sub/ultra-harmonics, or broadband noise as indicators of different types of cavitation activity, the relationship between bubble dynamics and acoustic emissions is not fully understood. It is this latter relationship that the present study seeks to explore.

There has been extensive use of high-speed imaging over the past four decades to capture bubble dynamics during both hydrodynamic and acoustic cavitation. This has been facilitated by the development of cameras, such as the Brandaris (Chin *et al.*, 2003), which was specifically designed with the goal of imaging bubble oscillations at ultrasound frequencies. More recent solid-state cameras (Kondo *et al.*, 2013) have further enabled research into bubble behavior at millions of frames per second with more convenient integration into experimental systems.

There has been similar growth in the use of both active and passive cavitation detection using acoustics, in particular, the development of two- and three-dimensional cavitation imaging or mapping (Gyöngy *et al.*, 2008; Haworth *et al.*, 2012; Salgaonkar *et al.*, 2009). There are relatively few studies, however, that simultaneously capture both acoustic emissions and optical footage of bubble dynamics. Rademeyer *et al.* captured both optical and acoustic scattering data from a range of different types of bubble using a

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flow-focusing device to create a steady stream of isolated bubbles (Rademeyer *et al.*, 2015). Optical scattering, however, only indicates how bubble volume varies as a function of time and cannot distinguish behavior such as microjetting. Sijl *et al.* (2011) used the Brandaris camera to capture footage of SonoVue contrast agent microbubbles simultaneously with the acoustic emissions at relatively low peak negative pressures (PNP) of 40 kPa. Under these conditions, both the radial amplitude and radiated pressure were found to be in good agreement with theoretical predictions. Where non-spherical behavior was observed, in the form of shape oscillations, this was not found to contribute significantly to the acoustic emissions. Sheeran *et al.* performed similar experiments using perfluorocarbon bubbles and droplets to demonstrate the difference in acoustic emissions produced by droplet vaporization vs microbubble oscillation (Sheeran *et al.*, 2014).

More recently, Song *et al.* used two high-speed cameras and a hydrophone to study the oscillations and acoustic emissions of SonoVue<sup>®</sup> (Bracco Diagnostics, Milan, Italy) microbubbles, finding that the acoustic emissions were dominated by band limited impulses generated during microbubble collapse (Song *et al.*, 2019). In their discussion, they questioned the validity of some commonly used categories for cavitation classification. The aim of the present study was to build on this previous work and further explore the relationship between bubble dynamics and acoustic emissions. Specifically, the following questions were addressed:

- Is there a correlation between the amplitude of microbubble oscillation and the amplitude and frequency content of the acoustic emissions?
- Are there clear transitions in microbubble behavior with varying acoustic excitation parameters and/or time, and are these mirrored by changes in the acoustic emissions?
- Do the characteristics of the acoustic emissions change when bubbles undergo significantly non-spherical oscillations, including phenomena such as microjetting or fragmentation?

- Is it possible to distinguish the emissions from single microbubbles from those of a cluster of equivalent volume?

## II. MATERIALS AND METHODS

### A. Experimental setup

A schematic of the experimental setup is shown in Fig. 1. The setup was designed to enable single microbubbles to be excited by an ultrasound pulse and their dynamics observed using a high-speed camera whilst their acoustic emissions were captured simultaneously by an ultrasound array. Microbubbles were drawn into a length of polyethylene tubing of 180  $\mu\text{m}$  inner diameter and 10  $\mu\text{m}$  wall thickness (Advanced Polymers, Salem, NH) using a low-pulsatility peristaltic pump (Minipulse Evolution, Gilson, Middleton, WI) at a constant rate of 4 ml/h, resulting in fully developed laminar flow in the region from which optical and acoustic data were acquired. The tubing was immersed in a water bath containing filtered and degassed water at the ambient temperature (21  $^{\circ}\text{C}$ ).

A 40 $\times$  objective lens with a numerical aperture of 0.8 and working distance 3 mm (LUMPLFLN 40XW Olympus, Hamburg, Germany) was focused on the mid-plane of the tubing and coupled to a high-speed camera (HPV-X2, Shimadzu, Kyoto, Japan). Illumination was provided by a high intensity light source [SOLIS-445  $^{\circ}\text{C}$ , Solis High-Power light-emitting diodes (LEDs), Thorlabs LTD., Ely, UK]. The instrumentation was triggered by detecting the light scattered by a bubble entering the objective focal volume. The scattered light was passed through a 480 nm laser line filter (FB440-10, Thorlabs) and was detected by a photomultiplier tube (PMT) (H10493, Hamamatsu Photonics Ltd, Japan).

A single element spherically focused ultrasound (FUS) transducer ( $f_0 = 0.5$  MHz centre frequency, H107, Sonic Concepts, Bothell, WA) was used to excite the microbubbles. The aperture and the geometric focus of the transducer were

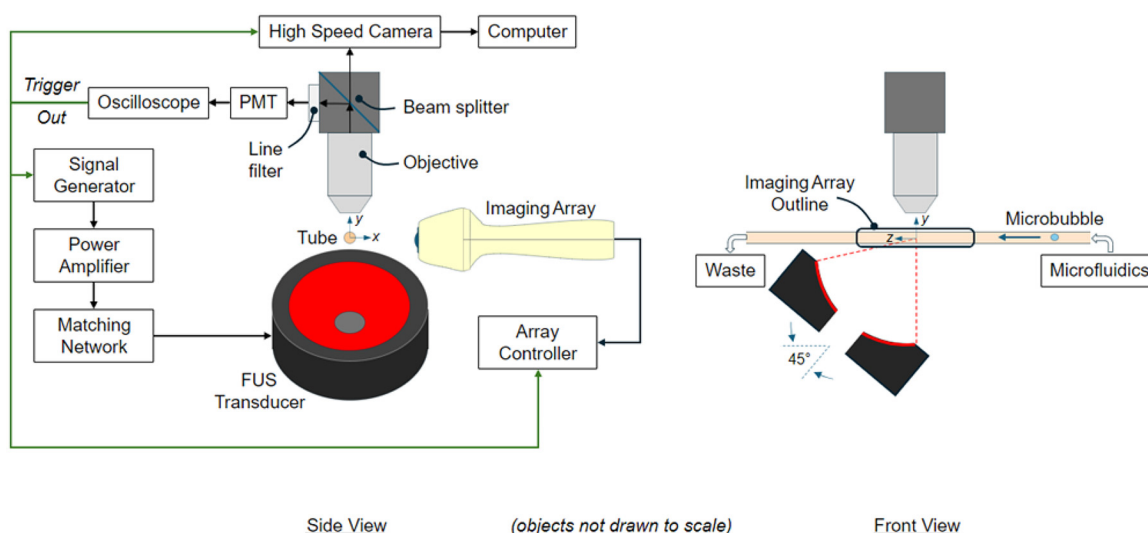


FIG. 1. (Color online) Schematic of the experimental setup for simultaneous optical and acoustic measurements of single microbubbles (side and front views).

64 mm and 63.2 mm, respectively, and the focal volume was aligned with that of the optical objective. The FUS transducer was excited using a waveform generator (33220 A, Agilent, Cheshire, UK) whose output was fed to a 300 W RF power amplifier (A-300, E&I Ltd, Rochester, NY) and sent to the FUS transducer via a 50 $\Omega$  matching network. The ultrasound exposure conditions used in the experiments were selected to coincide with those used in, e.g., blood brain barrier opening and drug delivery, as these are applications in which the distinction between different types of cavitation (i.e., inertial and non-inertial) is frequently discussed. The FUS transducer, which was angled to minimize the effects of reflection from the tank wall and/or air–water interface, was characterized using a calibrated needle hydrophone (HNC0400, Onda, Sunnyvale, CA) to confirm the mainlobe beam dimensions and the relationship between drive voltage and focal plane pressure. A diagnostic ultrasound imaging probe (L7-4 linear array, ATL Ultrasound, Bothell, WA) connected to a Verasonics Vantage ultrasound research system (Verasonics Inc., Redmond, WA) was positioned  $\sim$ 40 mm away from the target region to record acoustic emissions. The dimensions of the focal volume (assumed to be ellipsoidal and based on full width half maximum pressure) for the driving ultrasound transducer were 4.1 mm diameter and 25.2 mm length. The point spread function after beam forming (please see below) for the receiving probe was  $\sim$ 240  $\mu$ m diameter and 1.96 mm length. The optical field of view within the tubing was  $\sim$ 175  $\times$  180  $\mu$ m and the depth of field 0.57  $\mu$ m.

## B. Microbubble manufacturing

1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC) and 1,2-distearoyl-sn-glycero-3 phosphoethanol-amine-N-[methoxy(polyethylene glycol) –2000] (DSPE-PEG2000) were obtained from Avanti Polar Lipids (Alabaster, AL). Chloroform, phosphate buffered saline (PBS) and polyvinyl alcohol (PVA) were obtained from Sigma-Aldrich (Gillingham, Dorset, UK). Perfluorobutane (PFB) were obtained from FluoroMed, L.P. (Round Rock, TX). Lipid films were prepared by evaporation of chloroform from a 9:1 molar mixture of DSPC:DSPE-PEG2000. Lipids were resuspended in PBS at 4 mg/ml by stirring for 45 min on a hot plate at 80  $^{\circ}$ C.

Initially, microbubbles were prepared by probe sonication as follows: the lipid suspensions were placed in a 7 ml glass vial (Merck Life Science, UK Limited, UK) and sonicated for 2 min at 40% amplitude using a probe sonicator (Q125, 3 mm tip diameter, QSonica, LLC, Newtown, CT) positioned at the bottom of the liquid. Using a bespoke lid allowing access for the sonicator probe, the headspace was filled with PFB gas at a low flow rate for 5 min after which a second sonication at the liquid/gas interface for 30 s (80% amplitude power) was used to generate the microbubbles. The vessel was then sealed, transferred to an ice bath for 10 min, and the microbubbles used for experiments the same day. The microbubble suspension was diluted by a factor of 1:500 before use. In preliminary experiments, however, the

uniformity of sonicated microbubbles and also that of a commercial contrast agent (SonoVue, Bracco Diagnostics, Geneva, Switzerland) were found to be too low to obtain repeatable results. The sonicated bubbles were therefore only used for the observations of multiple bubble interactions.

To better control their size and rate of entry into the experimental system, microbubbles were also made using a microfluidic device. Flow-focusing microfluidic chips were fabricated following a previously published protocol (Seo *et al.*, 2010). Briefly, microfluidic devices were cast in polydimethylsiloxane (Sylgard 184, Dow Corning, Midland, MI) from a custom SU-8 mold. The device was then plasma bonded to a clean polydimethylsiloxane substrate and heated at 70  $^{\circ}$ C for 2 h. The final dimensions of the gas and liquid channels were 15 and 20  $\mu$ m wide, respectively, the nozzle was 5  $\mu$ m wide, and all channels were around 20  $\mu$ m height. The outer phase was a mixture of lipids (as described above, 4 mg/mL) and PVA (0.25 mg/mL) solution. The gas phase was PFB. The mean diameter of microbubbles produced by microfluidics was  $4.2 \pm 0.9 \mu$ m.

For the majority of the reported experiments, the microfluidic chip was directly coupled to the polyethylene tubing in the water tank. As above, the liquid flow rate was set with a low-pulsatility peristaltic pump (Minipulse Evolution, Gilson, Middleton, WI) at a flow rate of 4 ml/min h ( $\sim$ 3.5 cm/s mean velocity in tube), and the gas pressure was set with a two-stage mini pressure regulator (Sigma-Aldrich, Gillingham, Dorset, UK) at a gas pressure of  $\sim$ 70 kPa (10 psi). Under these conditions, a steady laminar flow could be maintained in the tubing and the microbubbles confined to the centre line of the tubing, i.e.,  $\sim$ 90  $\mu$ m away from the tubing walls to minimize boundary interactions at the start of each experiment; inevitably, bubble translation as the result of radiation forces was observed over the 50 cycle pulse (Garbin *et al.*, 2007).

## C. Experimental procedure

The experimental procedure was as follows: First, the foci of the microscope objective, focused transducer, linear ultrasound array, and light source were coaligned with the centre of the polyethylene tubing, by adjustment of the manual positioning stages on which they were mounted (model 443, Newport, Cheshire, UK). During this process, the tubing was filled with air to allow it to be used as an imaging/scattering target for the ultrasound transducers and to provide optical contrast for aligning the microscope objective. A measurement distance of 40 mm was chosen for the array as a compromise between cavitation signal to noise ratio and response complexity from element diffraction patterns.

Second, liquid was drawn into the tubing by activating the syringe pump containing microbubbles either from a reservoir of pre-made sonicated bubbles or from the microfluidic device. As a microbubble passed through the focus of the microscope, the change in light intensity was detected by the PMT and the resulting signal was used to trigger the waveform generator, high-speed camera, and ultrasound array controller.

Microbubbles were excited at 0.5 MHz with a 50-cycle (100  $\mu$ s) pulse and varying peak negative pressures. After a delay of 40  $\mu$ s to allow for propagation of the ultrasound pulse to the bubble, the high-speed camera recorded 256 frames at speeds of either  $1 \times 10^6$  frames per second (Mfps) with 200 ns of exposure time, to capture the motion of the bubble over the full 50 cycles of the driving pulse, or 5 Mfps to better resolve the bubble dynamics over 20 driving cycles. As discussed in Sec. IV B, however, it should be noted that neither the spatial nor temporal resolution of the system is sufficient to fully capture the bubble dynamics.

Digital images of  $400 \times 250$  pixels were recorded at an image resolution of  $0.7 \mu\text{m}/\text{pixel}$ , determined using a hemocytometer as a reference standard (Bright-Line, Hausser Scientific, Horsham, PA). The acoustic emissions from the microbubble were simultaneously received by the linear array for the full 50 cycles, the signal from each element was digitized at 25 MHz, and stored for offline analysis. Each experiment was repeated at least five times. Due to the highly irregular shape of the bubbles observed at higher PNP, the projected bubble cross sectional area (rather than diameter) was estimated from the optical images using MATLAB (The MathWorks, Natick, MA) (Schindelin *et al.*, 2012). The ratio of maximum to the initial projected area of the bubble (RMIA) was also calculated for each experiment.

### D. Acoustic array data processing

The array was calibrated as a function of frequency to correct for element-by element sensitivity and diffraction effects so that bubble emissions could be reported in device-independent physical units (Gray and Coussios, 2018). The recorded waveforms from each element were transformed into the frequency domain, corrected for the complex-valued element sensitivities, and retained for spectral analysis. They were also transformed back to the time domain and filtered using an 8th order 2–8 MHz Butterworth bandpass filter to minimize enhancement of noise outside the array response band for display and calculation of cavitation maps.

Passive acoustic mapping (PAM) was used to localize the microbubble in the imaging field of view and to estimate the microbubble's radiated acoustic energy. The robust Capon beamforming (RCB) algorithm (Coviello *et al.*, 2015) was used with the optimization control parameter (“epsilon”) set to unity based on prior studies with calibrated arrays (Gray and Coussios, 2022). To study the time-evolution of bubble activity during a pulse, array time domain data were steered to the bubble location estimated from robust Capon beamforming-passive acoustic mapping (RCB-PAM) and averaged over all 128 elements to provide an estimate of the acoustic emissions from the bubble. Doing so provides considerable background noise suppression relative to single element or hydrophone measurements made at the same centreline distance.

Spectrograms of these acoustic emissions data were calculated using a 20  $\mu$ s (10 FUS cycle) sliding Hamming window with a window overlap duration corresponding to the

inter-frame time of the video recording (1  $\mu$ s). The spectral features of the cavitation data were quantified by estimating the harmonic ( $f_{h,n} = nf_0 = \text{integer multiples of the FUS drive frequency}$ ), ultraharmonic ( $f_{u,n} = (n - 1/2)f_0 = \text{odd multiples of half the FUS drive frequency}$ , where  $f_{u,1} = f_0/2$  is the subharmonic), and “broadband” components of the calibrated and steered spectra. For the  $n$ th harmonic, the mean ( $\overline{PSD}_{h,n}$ ) and standard deviation ( $\sigma_{h,n}$ ) of the power spectral density (PSD) were calculated in the band defined by  $bw_n = f_{h,n} \pm \Delta bw/2$ . The summed power in this band ( $P_{h,n} = bw_n \times \overline{PSD}_{h,n}$ ) was classified as harmonic if the signal power  $PSD(nf_0) > \overline{PSD}_{h,n} + 2\sigma_{h,n}$ , and otherwise set to zero. Evaluations of ultraharmonics were carried out the same way, with  $\Delta bw$  for all calculations set to 70 kHz based on dataset review. The broadband portion of the signal power was found from  $P_{bb} = P_{total} - \sum_n (P_{h,n} + P_{u,n})$ , where  $P_{total}$  is the integrated PSD over the full calibration band of the receive array (2–8 MHz). The above procedure is intended to avoid underestimation of broadband contributions when no clear tonal response is detectable, and the use of  $\Delta bw$  rather than a single spectrum bin (typically 5 kHz) is intended to avoid overestimating broadband contributions when the harmonic and/or ultraharmonic responses have finite bandwidth.

## III. RESULTS

### A. Data collected for each experiment

An example of a dataset collected for a single microbubble driven at 0.5 MHz with a PNP of 0.6 MPa is shown in Fig. 2. Figure 2(a) shows a sequence of high-speed images capturing the dynamic behavior of the bubble at a rate of 1 Mfps. Figure 2(b) shows the time domain signal resulting from steering the L7-4 array to the bubble location and Fig. 2(c) shows the corresponding power spectral density (depicted in blue) along with the background noise (shown in red, measured in the time interval prior to the arrival of the first signal from the flow tube). The signal and noise spectra are shaped by the array response calibration and bandpass filter, leading to flattened broadband signal emissions in the calibration band and to accentuated levels between 5 and 8 MHz in the apparently non-acoustic background noise. Figure 2(d) shows the evolution of the frequency content with time as a spectrogram. Please note that the frame number ( $F$ ), rather than capture time, is shown for each image from the high-speed camera footage. This is due to the possibility of variable delays between the capture of the optical and acoustic signals. The reasons for these delays are discussed further in Sec. IV. All video footage is available as supplementary material.

### B. Effect of driving pressure on bubble dynamics and acoustic emissions

To investigate the effect of pressure on bubble dynamics, the PNP of the driving ultrasound burst was varied from 0.12 MPa to 1.2 MPa while the driving pulse centre frequency and pulse length were fixed at 0.5 MHz and 50

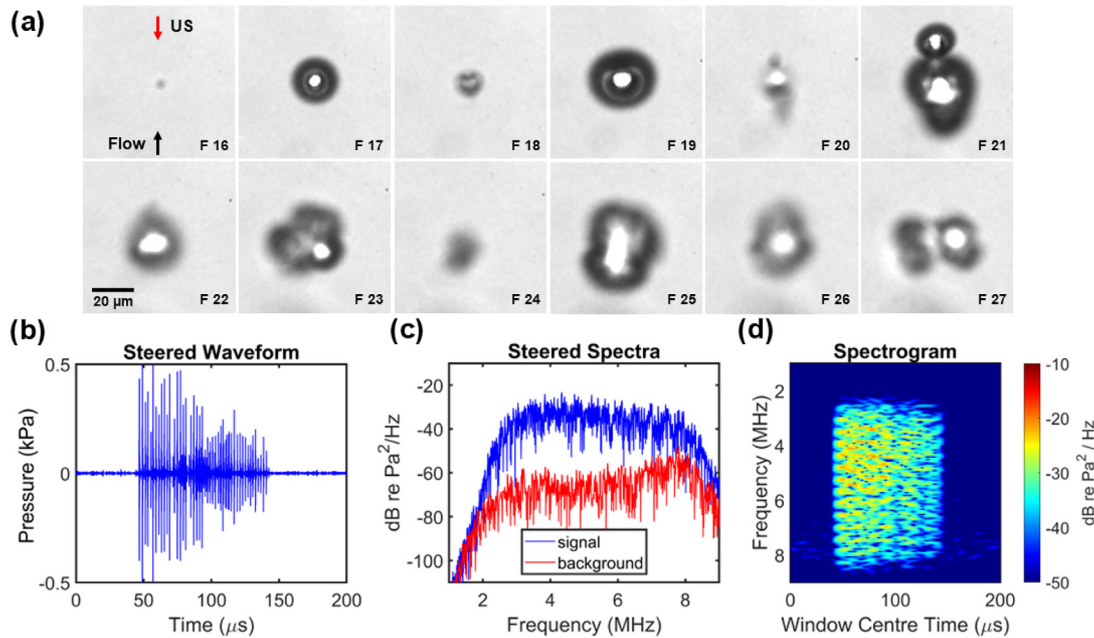


FIG. 2. (Color online) Example set of data acquired for each experiment. (a) A sequence of high-speed images capturing the dynamic behavior of a single microbubble under ultrasound exposure, (b) the beam-formed acoustic emissions captured from the oscillating microbubble, (c) the corresponding channel-averaged frequency spectra for the captured waveform, (d) their spectrogram. The centre frequency, pulse length, and PNP of the driving pulse were 0.5 MHz, 50 cycles, and 0.6 MPa, respectively. The directional indicators for flow and ultrasound from the driving transducer in panel (a) are denoted by black and red arrows, respectively. The receiving probe was positioned to the left of the optical images. Scale bar = 20  $\mu\text{m}$ . Please note that due to uncertainty in the capture time of the optical and acoustic data, frame numbers rather than times are shown in the high-speed images.

cycles, respectively. Figure 3 compares the projected area of the bubble as a function of time measured from the high-speed images (first column), the steered time traces (second column), corresponding power spectra (third column), and their spectrogram (fourth column) at different PNP. Please note that, due to the large changes in projected area at higher PNP, the vertical axis has been scaled accordingly.

At a PNP of 0.12 MPa, the high-speed camera images show that the microbubble underwent small amplitude oscillations and, whilst translation towards the tube wall was observed [presumably due to radiation force(s)], it retained its initial shape (Video S1) in the visible imaging plane, and this was captured by the periodic oscillations in the pixel count [Fig. 3(a)]. In the PSD and spectrograms, there were clear harmonic acoustic emissions that existed for the duration of the excitation, resulting in sharp harmonic peaks ( $\sim 15$  kHz wide at  $-10$  dB re peak). At this pressure, there was not a substantial change of the broadband spectra above the background noise. However, upon increasing the PNP to 0.24 MPa [Fig. 3(b); Video S2], the bubble could be seen for 40 frames (during which it appeared to retain a spherical shape) and then vanished; similarly, the emissions were present for the first  $\sim 40$   $\mu\text{s}$  and also disappeared. The power spectra exhibit broader peaks due to the truncation in the time domain and the spectral shifts. In addition to the harmonic components, a broadband level  $\sim 20$  dB above background was observed. Similar results were obtained in the majority of the experiments performed at this pressure, i.e., both the optical and acoustic data suggested that bubble destruction occurred more quickly than at higher or lower pressures.

When the PNP was increased to 0.6 MPa [Fig. 3(c); Video S3], the bubble lost its spherical shape within a single frame of the video capture and, in many cases, fragmentation resulted in multiple bubbles in the field of view. The number of pixels that the bubble took up was  $\sim 5$  times larger than at 0.24 MPa. Acoustic emissions were detected for the entire duration of the excitation, and apart from the first  $\sim 10$  frames, there did not appear to be any clear harmonics. The broadband emissions were  $\sim 40$  dB above background. At 1.2 MPa PNP, a similar transition to a complex bubble field was observed [Fig. 3(d); Video S4]. Again, harmonics were present in the spectrogram for the first  $\sim 10$  frames but then disappeared and could not be subsequently identified in the PSD. Notably, discrete ultraharmonics were not detected in the spectra or spectrograms. There was also no clear correlation between bubble translation and change in spectral content across all of the datasets recorded.

Figure 4(a) shows how the magnitude of the harmonic and broadband content of the acoustic emissions varied with increasing PNP. In Fig. 4(b), increase in the broadband signal with PNP is plotted together with the mean average and maximum bubble projected area. Figure 4(c) shows how the bubble RMIA varied with PNP.

### C. Effect of different phenomena on acoustic emissions

#### 1. Microjets

Although the experimental setup was designed to avoid bubbles coming into close proximity with either each other

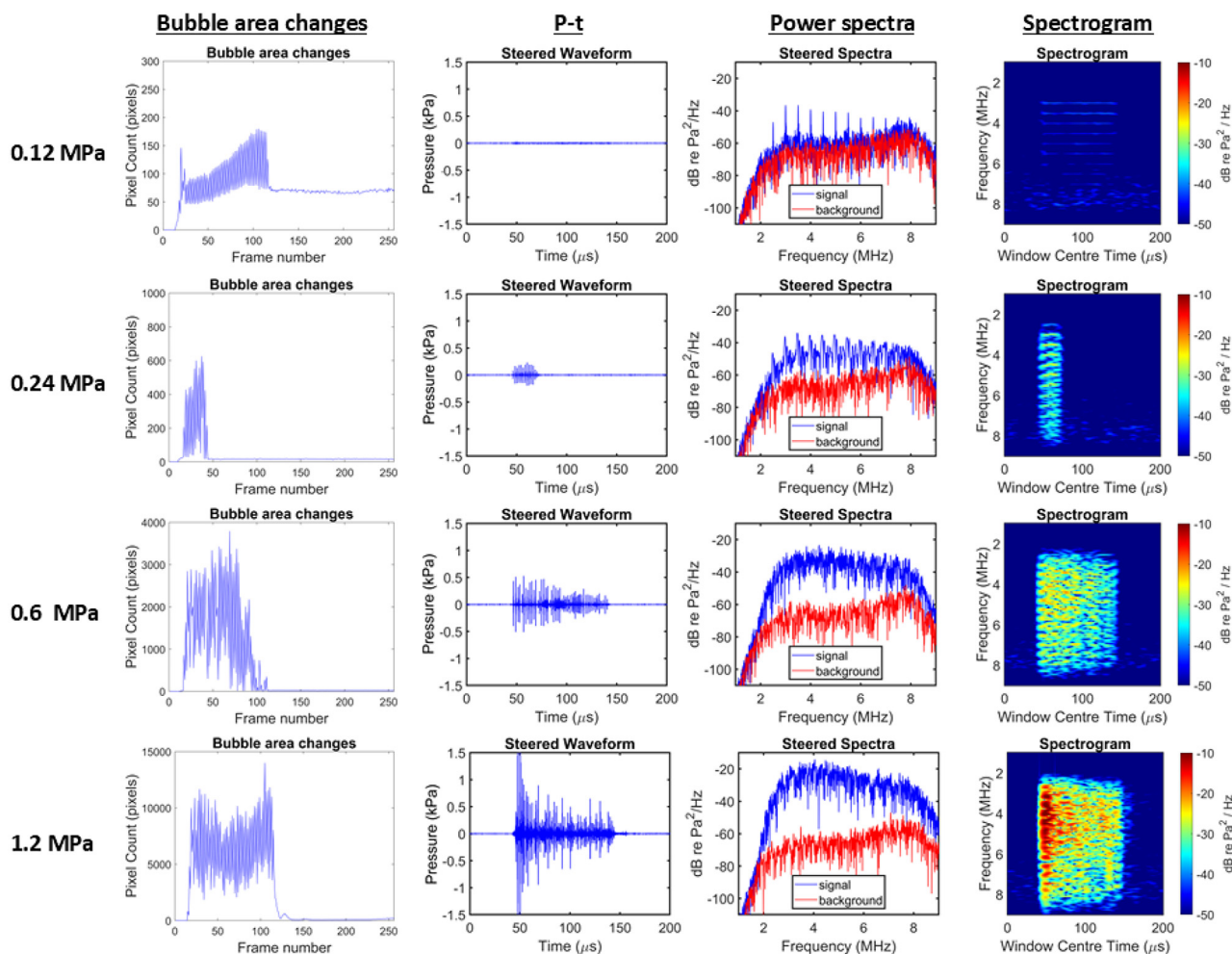


FIG. 3. (Color online) Representative datasets for single microbubbles excited by a 50 cycle ultrasound pulse having a centre frequency of 0.5 MHz and different peak negative pressures. The first column shows how the projected area of the bubble changed with time as detected with the high-speed camera. The second column shows the beam-formed acoustic emissions captured during bubble oscillation P-t. The third column shows the corresponding frequency spectra for the acoustic emissions and the fourth column shows their spectrogram.

or the walls of the tubing, microjet formation was still observed in some of the experiments. An example is shown in Fig. 5 for a bubble driven at 0.6 MPa PNP. In the high-speed video footage [Fig. 5(f); video S5], microjet formation was observed whilst the bubble was collapsing as would be

expected (F18). Subsequently the bubble started to expand and interestingly, retained its involution shape (F19). It then underwent fragmentation, followed by coalescence of the daughter bubbles (F21-27). There were substantial changes in the amplitude of the acoustic emissions over this time

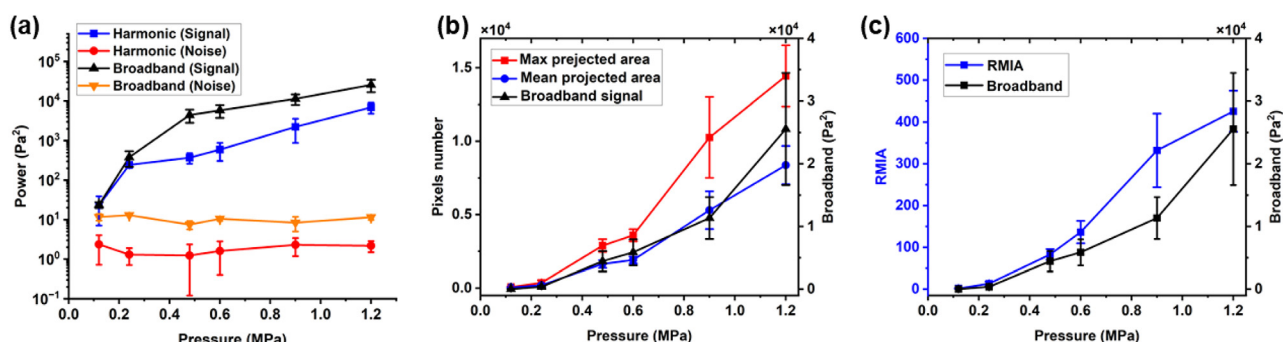


FIG. 4. (Color online) The effect of driving pressure on the amplitude and frequency content of acoustic emissions and amplitude of bubble oscillation. (a) The harmonic and broadband content of the acoustic emissions detected from individual microbubbles and corresponding background noise as a function of PNP, (b) comparison between the mean and maximum bubble projected area from the high-speed video images and broadband signal measured by the acoustic emissions as a function of PNP, (c) comparison between the ratio of maximum to initial bubble projected area and broadband signal as a function of PNP. The mean average and standard deviation for at least five bubbles are shown for each PNP.

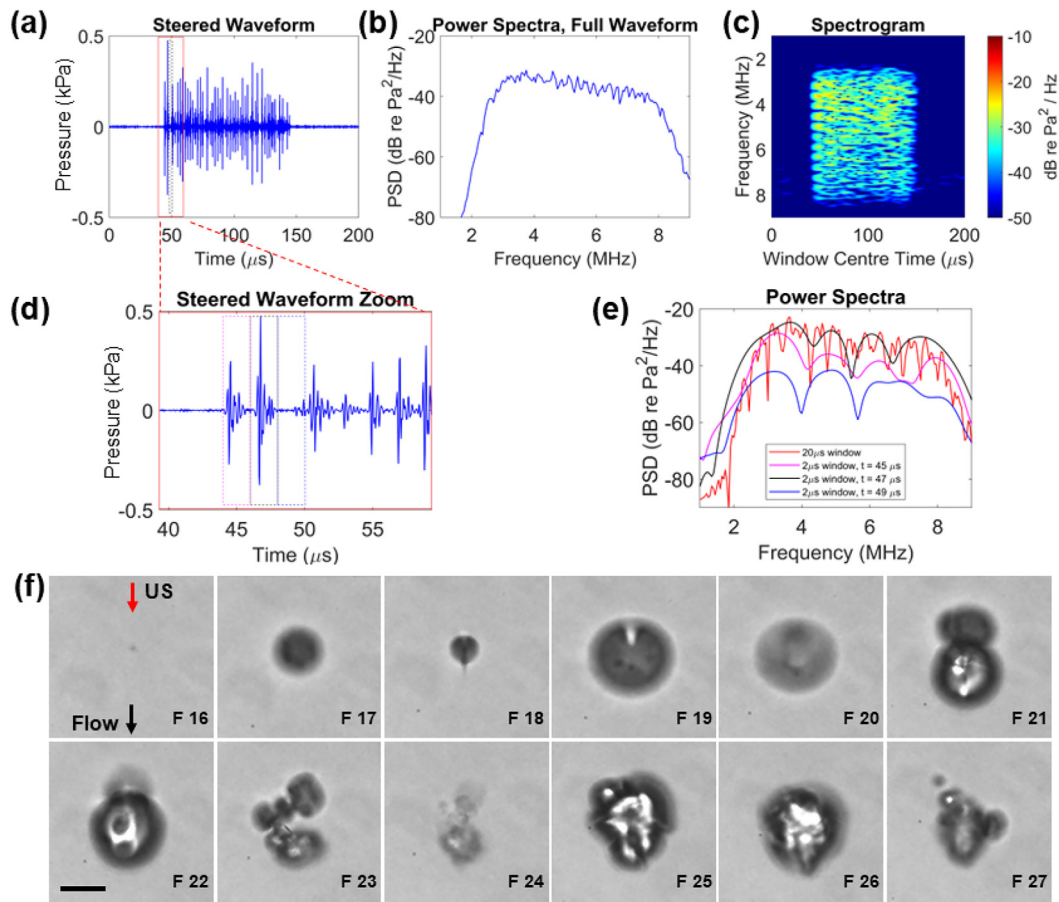


FIG. 5. (Color online) Results for a single microbubble excited at  $PNP = 0.6$  MPa undergoing microjetting (video 5). (a) Acoustic emissions captured as a function of time, (b) the corresponding frequency spectrum, (c) spectrogram, (d) a subset of the captured acoustic emissions with (e) frequency spectra for a series of  $2 \mu\text{s}$  windows covering the period of microjetting, (f) the sequence of high-speed camera images for the same period. The directional indicators for flow and ultrasound are shown by black and red arrows, respectively. The probe was positioned to the left of the optical images. Scale bar =  $20 \mu\text{m}$ .

[Figs. 5(a) and 5(d)]. The frames shown in Fig. 5(f) correspond to the section of the recorded acoustic signal shown in Fig. 5(d). As mentioned above, the precise location of each optical video frame with respect to the acoustic time trace is subject to an uncertainty of  $\sim 0.5 \mu\text{s}$  (cf. also Sec. IV). There was, however, no significant change in the range of frequencies present in the signal over the time during which microjetting occurred when this was analysed over consecutive  $2 \mu\text{s}$  windows [Fig. 5(e)] (cf. also supplementary material).

## 2. Fragmentation

Bubble fragmentation was also observed at  $0.6$  MPa PNP, as shown in Fig. 6 and Video S6. Based on the optical images [Fig. 6(f)], there was a fragmentation event between frames 20 and 32. The amplitude of the acoustic emissions recorded over this time period again changed substantially [Fig. 6(d)]. As shown in Fig. 6(e), in this case, there was a clear change in the frequency content of the signal over consecutive  $2 \mu\text{s}$  windows, with a large drop in the amplitude of the higher frequency components. This change in signal amplitude above  $6$  MHz is also clearly visible in the spectrogram [Fig. 6(c)]. This information is lost in the time-integrated spectra for the whole recorded signal [Fig. 6(b)]

and for the  $20 \mu\text{s}$  window selected in Fig. 6(d). It is important to note that the shape of the bubble in the optical images suggests that microjetting may have been occurring simultaneously with the fragmentation event. As it was only possible to capture a single optical plane with the experimental setup, however, this could not be verified.

## 3. Coalescence

Under the same excitation conditions, both fragmentation and coalescence of bubbles were seen in some cases as shown in Fig. 7 and Video S7. As for fragmentation, there was again some evidence of a reduction in the amplitude of the measured acoustic emissions compared with earlier in the pulse [Fig. 7(d)] and a change in the frequency content [Fig. 7(e)], although not all of the high frequency components were attenuated in this example as was the case in Fig. 6.

## D. Single vs multiple bubbles

Since biomedical applications of cavitation typically involve populations of bubbles, a suspension of pre-made bubbles was also imaged under the same excitation conditions as the individual microfluidic-generated microbubbles. The concentration of pre-made bubbles pulled into the

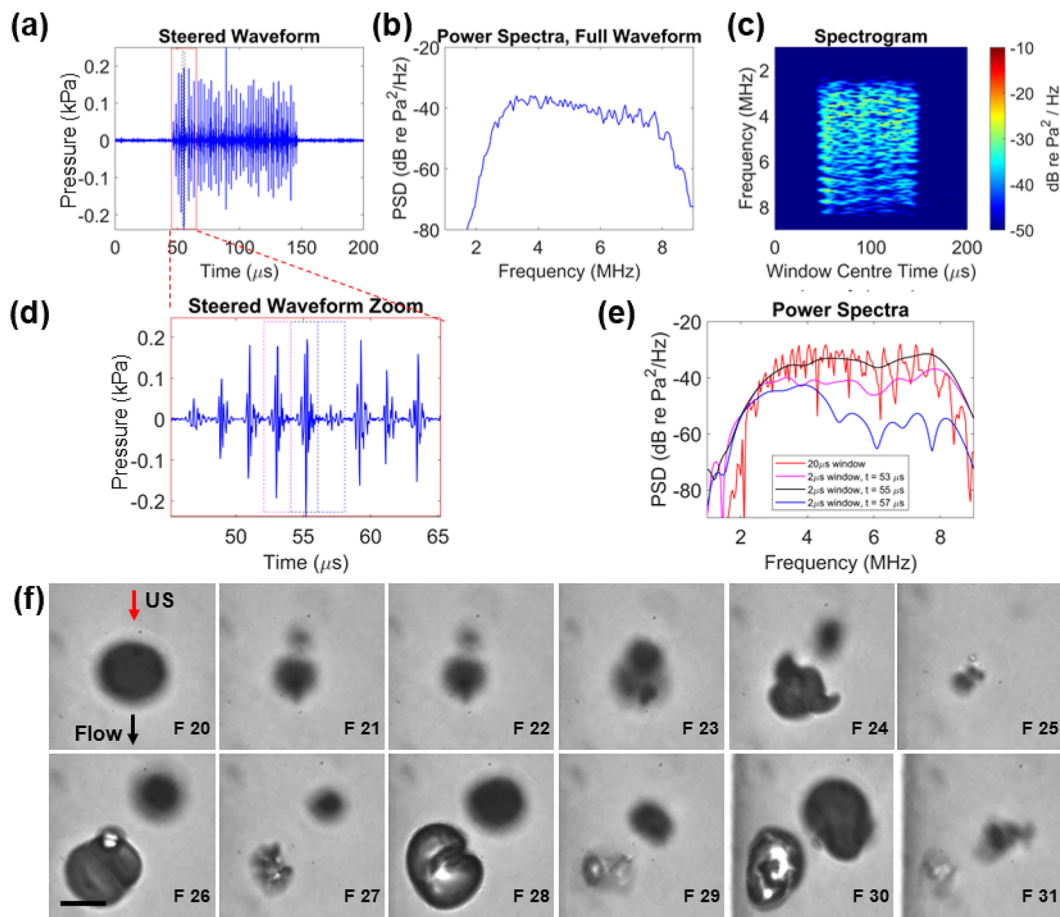


FIG. 6. (Color online) Results for a single microbubble excited at PNP = 0.6 MPa undergoing fragmentation (video 6). (a) Acoustic emissions captured as a function of time, (b) the corresponding frequency spectrum, (c) spectrogram, (d) a subset of the captured acoustic emissions with (e) frequency spectra for a series of 2  $\mu$ s windows covering the period of fragmentation, (f) the sequence of high-speed camera images for the same period. The directional indicators for flow and ultrasound are shown by black and red arrows, respectively. The probe was positioned to the left of the optical images. Scale bar = 20  $\mu$ m.

tubing was  $\sim 10^6$  particles/mL. Supplementary Video S8 shows the resulting bubble clusters, a frame from which is shown in Fig. 8(a). As would be expected, the amplitude of the acoustic emissions was greater than for single bubbles of equivalent size [e.g., Fig. 8(b) vs Fig. 2(b)] but there were no obvious differences in the corresponding frequency content [e.g., Fig. 8(c) vs Fig. 2(c)] of the acoustic emissions integrated over the whole pulse as would typically be used for treatment monitoring. With increasing numbers of bubbles, there was also increased incidence of jetting, fragmentation, and coalescence as would be expected due to increased probability of bubble–bubble interaction. The spectrogram [Fig. 8(d)] indicates that again, the amplitude of higher frequency components diminished over the course of the pulse, but it was not possible to relate the changes to specific phenomena.

## IV. DISCUSSION

### A. Implications for therapy monitoring

The experiments in this study were motivated by applications to monitoring ultrasound therapy. The experiments

were conducted under controlled conditions with far less variability than would occur to a bubble *in vivo*. This was to enable investigation of the relationship between bubble dynamics and acoustic emissions with as few confounding factors as possible. Even under these conditions, however, there was seen to be a significant degree of variability in the behavior of nominally identical microbubbles as observed by high-speed imaging [cf. error bars in Fig. 4(b)]. This is consistent with earlier reports of commercial contrast agent bubbles (Rademeyer *et al.*, 2015) and is perhaps unsurprising, given the well-known stochastic nature of nonlinear bubble dynamics. What is perhaps surprising is that the common features of acoustic emissions, e.g., harmonics, and broadband, did not appear to correlate strongly with the microbubble behavior. As shown in Figs. 3 and 4, there is a positive correlation between the amplitude of microbubble oscillation and the energy of the broadband content of the acoustic emissions. There are, however, some interesting points to note:

- The frequency content of the acoustic emissions calculated using the whole duration of the 50 cycle driving pulse was not substantially different for the very different forms of bubble behavior observed, i.e. spherical

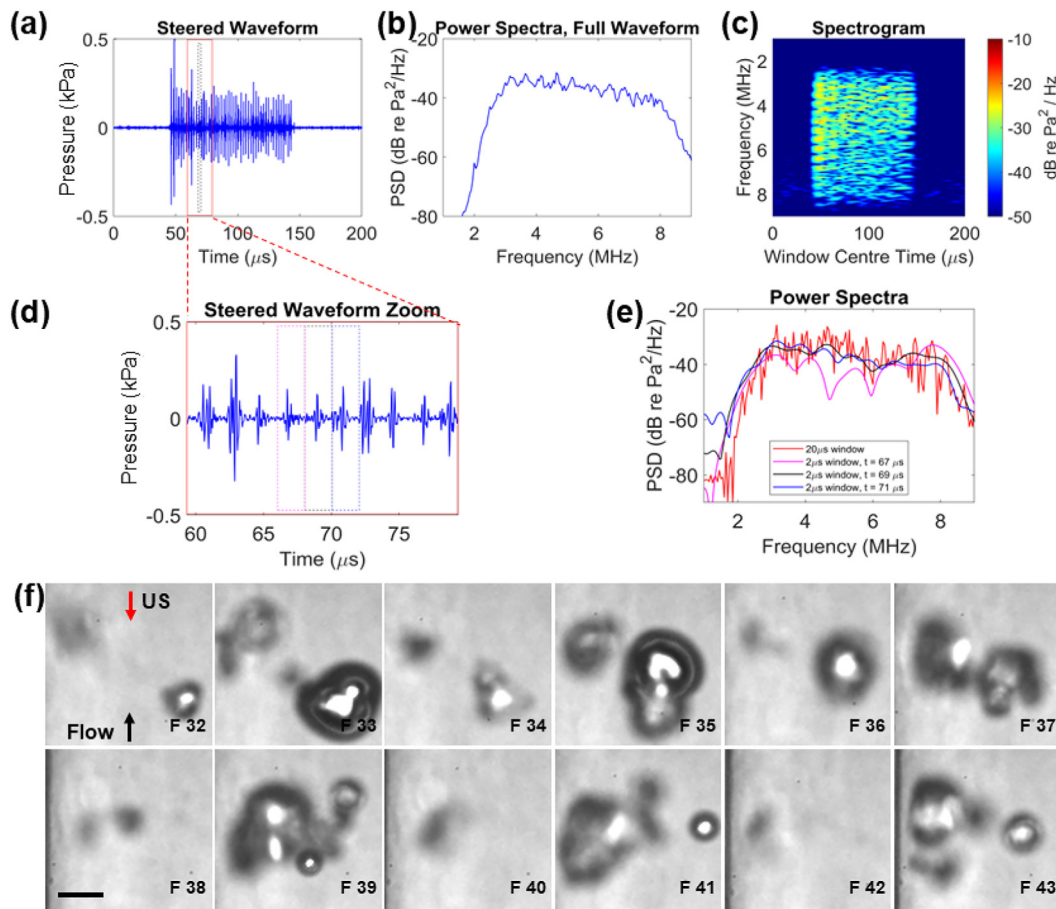


FIG. 7. (Color online) Results for a single microbubble excited at  $PNP = 0.6$  MPa undergoing coalescence (video 7). (a) Acoustic emissions captured as a function of time, (b) the corresponding frequency spectrum, (c) spectrogram, (d) a subset of the captured acoustic emissions with (e) frequency spectra for a series of  $2 \mu s$  windows covering the period of coalescence, (f) the sequence of high-speed camera images for the same period. The directional indicators for flow and ultrasound are shown by black and red arrows, respectively. The probe was positioned to the left of the optical images. Scale bar =  $20 \mu m$ .

oscillation, fragmentation, coalescence, and microjetting [Figs. 5(b), 6(b), and 7(b)]. In all cases, similar levels of broadband noise were detected. These phenomena could, however, have very different effects upon tissue. Nor was it possible to distinguish the behavior of individual microbubbles from that of a cloud from the acoustic emissions alone [Fig. 8(c)].

- The temporal evolution of the acoustic emissions [Figs. 5(b), 6(b), and 7(c)] did in some cases reflect changes in the bubble dynamics observed with the high-speed camera. Specifically, there was a loss in higher frequency components over time. This may have been due to bubble fragmentation and the smaller amplitude, less nonlinear behavior of the resulting daughter bubbles.
- An important source of harmonic signals in the acoustic emissions is the periodicity of the bubble collapses and the corresponding impulse response of the receiving transducer. This effect has been reported in previous studies as a potential source of sub-harmonic components as well (Gyöngy and Coussios, 2010; Song *et al.*, 2019). Analysis in the supplementary material shows that the time between collapses was relatively consistent for driving amplitudes of 0.12 and 0.24 MPa (within  $0.2 \mu s$ ) and the

corresponding PAM data showed multiple harmonics. For the other datasets, the time between collapses could vary by  $1 \mu s$  or more and harmonics were consequently not detected in the spectra.

- The evolution of single bubble emissions from harmonic to broadband often occurred without exciting a fractional harmonic state observable either in the frequency domain or in the time domain. Indeed subharmonics (and their multiples) were only detected for two cases, Figs. 5 and 6, and only in the Fourier transform of the envelope of the PAM signal (see Fig. S6).
- The non-spherical components of bubble oscillations did not appear to contribute substantially to the acoustic emissions, in that the spectra were not readily distinguishable from spherical oscillations (e.g. Fig. 6). This contradicts the theoretical predictions of Longuet-Higgins (Longuet-Higgins, 1989) but is consistent with Sijl *et al.* (2011) and with the expectation that multipolar motions of acoustically small radiators should have minimal volume velocity and consequently minimal acoustic emissions (Pierce, 2019). It is possible that by employing a larger observation aperture, measurements of the directional radiation could identify fragmenting, coalescing, or jetting bubbles.

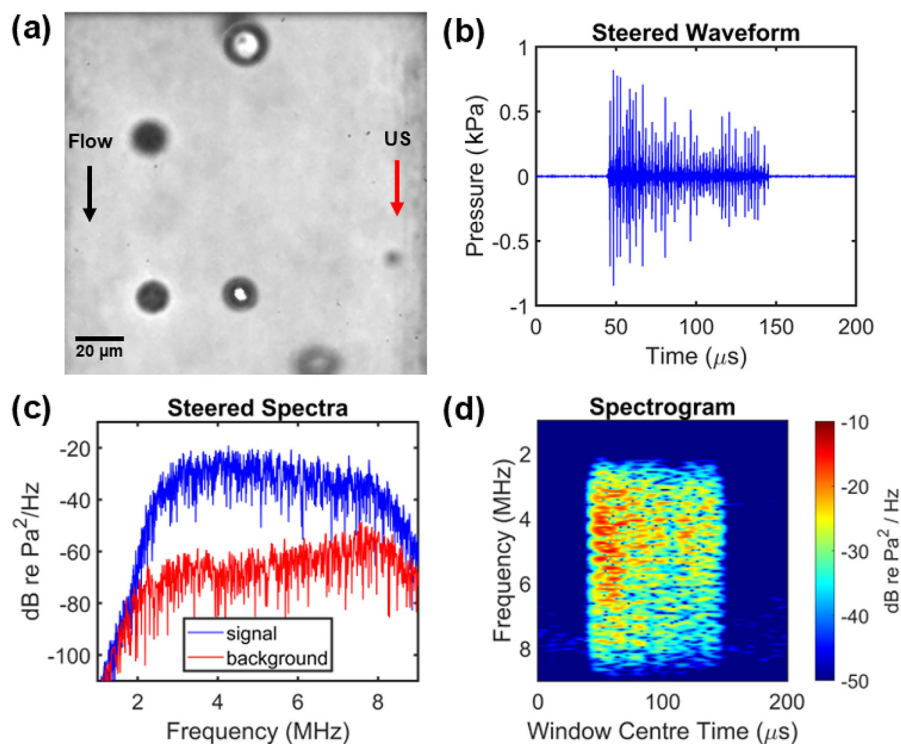


FIG. 8. (Color online) Results for multiple bubbles excited at  $PNP = 0.6$  MPa (video 8). (a) Representative high-speed camera image, (b) acoustic emissions captured as a function of time, (d) the corresponding frequency spectrum, (d) spectrogram. The directional indicators for flow and ultrasound are shown by black and red arrows, respectively. The probe was positioned to the left of the optical images. Scale bar =  $20\ \mu\text{m}$ .

Using larger apertures, however, is challenging clinically. Moreover, even if complex shape oscillations lead to accentuated or attenuated emissions for a single bubble for a single cycle, these changes will likely be masked by the presence of additional bubbles undergoing low(er) order vibrational modes and/or when observations are integrated over multiple fundamental cycles [cf. Fig. 6(e)].

- The shapes of the curves in Fig. 4(c) obtained from the optical and acoustic data with increasing PNP are different. The RMIA shows a sharper transition in its gradient that is consistent with expectations from theoretical modelling of the onset of inertially dominated collapse (Flynn, 1975a,b). There is, however, no distinct threshold for the onset of broadband emissions, which increase relatively smoothly with driving peak negative pressure. This is consistent with the findings of earlier studies of air bubbles in water, which also reported that the measured level of broadband noise grew gradually with increasing driving pressure (Neppiras, 1980).
- At  $0.24$  MPa PNP, the RMIA was  $13 \pm 5$ , [Fig. 4(c)] i.e., the maximum equivalent radius was greater than 2.3 times the initial bubble radius, a widely used definition of “inertial” cavitation (Neppiras and Noltingk, 1951) but interestingly, the change in gradient of the RMIA with PNP occurred at a higher pressure. It was not possible to conclusively test the related criterion for inertial cavitation that the bubble wall velocity exceeds the speed of sound in the liquid, since, even at a higher frame rate (Video S9), the uncertainty in the measured bubble wall velocity was substantial during the most rapid period of bubble collapse (Fig. S10; Table S1). The velocity criterion, however, derives from the same analysis as the

radial expansion threshold (Neppiras and Noltingk, 1951). It has, moreover, also been proposed as a condition for the emission of broadband noise (Apfel, 1981) whereas numerical simulations of bubble behavior using a modified Herring equation (Morgan, 2000) (cf. supplementary material: “modelling of bubble oscillations and acoustic emissions”) demonstrate that oscillating bubbles can emit broadband signals when driven at amplitudes below those at which the speed of the bubble wall appears to exceed the speed of sound in the surrounding liquid (Fig. 9).

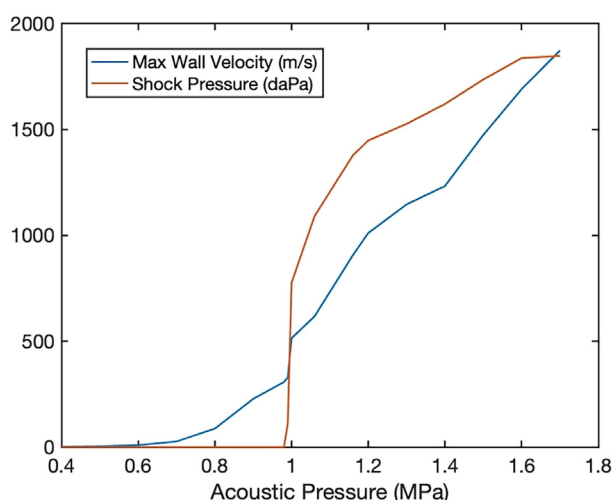


FIG. 9. (Color online) Predicted shock pressure (orange curve) and maximum wall velocity (blue curve) as a function of PNP. The predicted shock pressure is zero up to  $PNP = 0.98$  MPa after which a shock wave results from the nonlinear distortion and would be detected as broadband noise. The bubble wall velocity does become supersonic relative to the liquid until  $PNP = 1.53$  MPa.

- At a driving PNP of 0.24 MPa, there was a much more rapid loss of visible bubbles and detectable acoustic emissions than at higher or lower PNPs (Fig. 3). This was seen consistently between different experiments. This could potentially be because 0.24 MPa was a sufficiently high pressure to cause bubble disruption, but insufficiently high to promote rectified diffusion and/or nucleation of air bubbles from the surrounding liquid. In contrast, at higher PNP, either or both of these phenomena could have occurred. Further experiments to quantify the incidence of different mechanisms of bubble destruction and with different levels and/or types of dissolved gas would be required.

With regard to the specific questions posed at the start of the paper, the following answers can now be offered:

- *Is there a correlation between the amplitude of microbubble oscillation and the amplitude and frequency content of the acoustic emissions?* - Yes, but only a qualitative one.
- *Are there clear transitions in microbubble behavior with varying acoustic excitation parameters and/or time, and are these mirrored by changes in the acoustic emissions?* - No.
- *Do the characteristics of the acoustic emissions change when bubbles undergo significantly non-spherical oscillations, including phenomena such as microjetting or fragmentation?* - Not in such a way that the changes in behavior could be inferred reliably from the acoustic emissions.
- *Is it possible to distinguish the emissions from single microbubbles from those of a cluster of equivalent volume?* - No, not with the configuration deployed in these experiments.

In summary, the observations are supportive of the assertion by Song *et al.* that categories commonly used in biomedical applications, such as stable and inertial cavitation, fail to even qualitatively represent the diversity of cavitation dynamics. Crucially, moreover, there is no simple correlation between different types of bubble behavior and the frequency content of the acoustic emissions. Indicators, such as the onset of broadband noise, or the appearance of ultra-harmonics are not appropriate for use as universal thresholds in therapeutic ultrasound. The feasibility of defining quantitative criteria for use in treatment monitoring requires further research and such criteria will likely be application specific, i.e., through relating a particular bioeffect to the relevant bubble dynamics and determining whether this can be inferred from acoustic emissions. These considerations are becoming increasingly important with the growing demand for a cavitation “dose” to be defined for therapeutic applications.

## B. Limitations of the study

Confinement of the microbubble flow by the polyethylene tubing and the use of the microfluidic device reduced the risk that the acoustic emissions were being generated by

bubbles out of the optical field of view but could not eliminate it. Where the PAM images indicated another bubble was present, these datasets were not used for single bubble analysis. As already discussed, there was a trade-off that had to be made in selecting the frame rate and time over which the optical and acoustical signals were acquired. To the best of the authors’ knowledge, with a single commercially available high-speed camera operating in the Mfps range, the maximum number of frames that can be captured is currently 256 frames (Cordin, Salt Lake City, UT and Shimadzu Corporation, Kyoto, Japan cameras). As has been shown, using multiple cameras with interleaved frames can improve temporal resolution (Li *et al.*, 2023), potentially enabling features of the bubble oscillations to be captured at higher frame rates than in this study. Even if four cameras were coupled together (the maximum likely to be practically feasible with commercially available beam splitters), this would still only enable bubble dynamics to be captured for a fraction of a typical therapeutic pulse. It is also likely that improved spatial resolution would be required and this is likely to remain the most challenging limitation for studying microbubble dynamics. This does impose a limit on the confidence with which the bubble radius and, in particular, radial velocity can be estimated during compression. Quantitative analysis of this uncertainty is included in the supplementary material.

As mentioned above, an allied limitation is the uncertainty in the acquisition timing of the video and ultrasound systems, despite them sharing a common trigger. Sound speed was estimated based on temperature, and with an uncertainty of 0.5 °C around a baseline temperature of 20.0 °C, the timing uncertainty for the ultrasound array over the full propagation distance from FUS source to flow tube to ultrasound array is estimated to be  $\tau_p = \pm 0.072$  ms. The timing uncertainty for data capture by the array controller is based on its sampling rate, and per manufacturer specifications,  $\tau_a = \pm 0.16$  ms for these experiments. The high-speed camera acquisition timing uncertainty relative to the trigger is  $\tau_c = \pm 0.05$   $\mu$ s, so the typical estimated uncertainty in video and ultrasound system relative timing is  $\sqrt{\tau_p^2 + \tau_a^2 + \tau_c^2} \sim 0.18$   $\mu$ s. There was, however, an additional source of uncertainty arising from the particular camera model used in these experiments which meant that in some cases, there was a delay of 1 frame (i.e., 0.5  $\mu$ s) between the optical and acoustic recordings. We note that uncertainty analysis of camera-acoustic emission timing is rarely reported in the literature, but it may have important implications for data interpretation.

As above, theoretically the diameter of the polyethylene tubing was sufficiently large to have a minimal impact on microbubble oscillations. It is unlikely however, that this was a valid assumption for the larger amplitude oscillations and/or after the first few cycles as acoustic radiation forces were clearly observed to cause bubbles to move towards the tubing walls. This will have almost certainly increase the incidence of nonspherical behavior. In previous experiments, it has been shown that a flow-focusing device can be used to perform similar experiments without the need for

flow confinement (Li *et al.*, 2023; Rademeyer *et al.*, 2015) It was, however, extremely difficult to avoid bubble-bubble interactions in the previous set up and it was therefore not used for this study.

Only a single bubble size was investigated in this study. This was both due to the volume of experiments required for even a single bubble size, and the potential that both size and coating properties would be altered by using different microfluidic chips thereby rendering fair comparisons impossible. Based on the literature, it would be expected that the amplitude of oscillation exhibited by a bubble at a given pressure and the magnitude of its translation would be affected by its initial size. However, within the range of bubble sizes relevant for biomedical applications, however (i.e., 0.5–10  $\mu\text{m}$ ) similar qualitative results would be expected.

PAM algorithms are designed to estimate the power or energy emitted from a monopole in a pre-defined set of spatial points. As such, they are not suited for detection of higher order bubble vibration modes that would have suppressed volume velocity and/or enhanced directivity unless these modes were accompanied with distinct spectral signatures. It should be noted, however, that no such signatures were detected in this study when calculating spectra over the full emission duration. Individual impulse spectra did on occasion show potentially identifiable features (cf. Fig. 6), but more work is needed to assess the reliability and causality of these short window features, and to further assess the feasibility and practical value of short window measures of cavitation activity. Spectral components above the driving frequency were the primary interest for this study. Similar experiments focusing on subharmonic emissions will be reported separately. The relationship between different types of bubble behavior and bioeffects was also not addressed in this study but will be essential for the development of appropriate safety indices.

## V. CONCLUSIONS

The aim of this study was to investigate the degree of correlation between microbubble dynamics and the acoustic emissions that they produce. High-speed video footage of microbubbles driven at different acoustic pressures was captured simultaneously with their acoustic emissions. As expected, the amplitude of microbubble oscillation increased with increasing PNP. Microbubble behavior also became increasingly unpredictable, with phenomena such as microjetting and fragmentation being frequently observed. The amplitude of the acoustic emissions also increased with PNP, thus confirming the utility of acoustic emissions monitoring during ultrasound mediated therapy, to verify the presence of bubbles at a target site and as a qualitative measure of cavitation vigor. Analysis of the spectral content, however, suggested that the use of spectral characteristics such as broadband noise could not be relied upon as indicators of specific types of bubble behavior. For example, it was not possible to distinguish spherical bubble oscillations from micro-jetting or bubble fragmentation or coalescence

from the frequency spectra calculated over the full emission waveform duration. Only when shorter integration windows were used to analyze the temporal evolution of the frequency content were some features detectable that could potentially be used as indicators of bubble fragmentation. Nor was there evidence to support the use of thresholds such as  $R_{max}/R_o > 2$  or  $\dot{R} > c_L$  being indicative of a transition from one type of bubble behavior to another. This uncertainty should be considered carefully when defining cavitation monitoring protocols or “doses” for a particular therapeutic application. Binary categories such as stable and inertial cavitation, or the use of spectral characteristics to infer these, should be avoided.

## SUPPLEMENTARY MATERIAL

See the supplementary material for high-speed camera videos; alternative methods of analysis for the temporal evolution of spectral content; illustration of harmonic generation by rapid bubble collapse; and numerical modelling of bubble oscillations and generation of acoustic emissions.

## ACKNOWLEDGMENTS

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## AUTHOR DECLARATIONS

### Conflict of Interest

Constantin Coussios is a director of Oxsonics Ltd. Eleanor Stride is a director of SonoTarg Ltd.

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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